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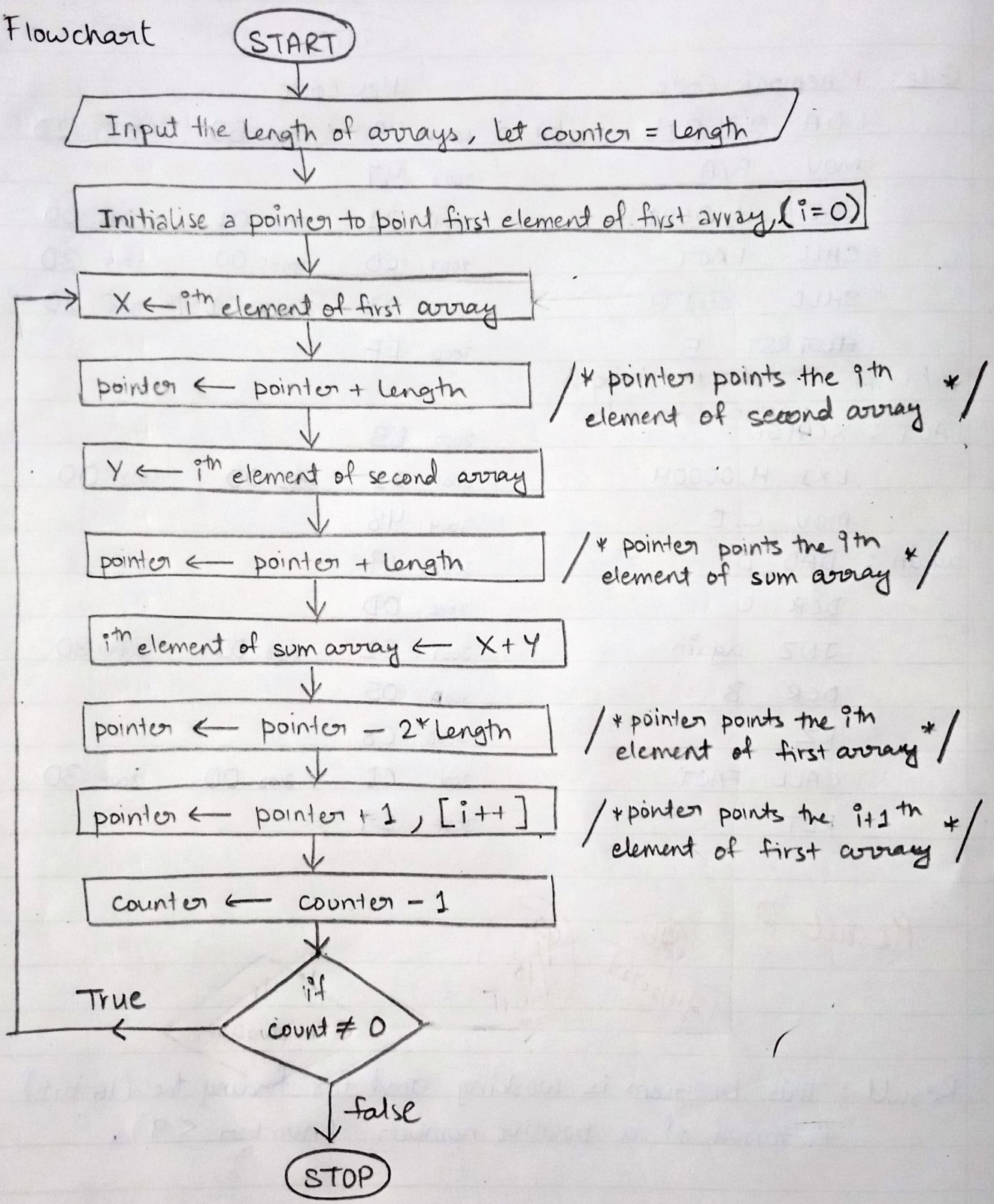
- Aim: Write a program in assembly language of 8085 to calculate and store the sum of two arrays of N 8bit numbers stored at consecutive memory locations.

- Algorithm:

(i) Input the size of the arrays, say N & M (M to control loop count).
(ii) Initialise a pointer to point first element of first array.
(iii) Store the ~~element~~ i^{th} element of first array in a variable, say X.
(iv) Increment the pointer by the size of the arrays to make it point the ith element of second array.
(v) store the ith element of second array in a variable, say Y.
(vi) add X & Y and store the sum in X i.e., $X \leftarrow X+Y$
(vii) Increment the pointer by the size of the arrays to make it point the ith element of third (sum) array.
(viii) Store the ith sum (X) in this location.
(ix) Decrement the pointer by the size of the array twice, to make it point the ith element of the first array.
(x) Increment the pointer by 1, to make it point $i+1^{th}$ element of the first array.
(xi) Decrement M by 1 as one loop iteration is over.
(xii) If M is not zero, jump to step (iii).
(xiii) If M is zero, program ends.

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• Flowchart


START
Input the length of arrays, Let counter = Length
Initialise a pointer to point first element of first array, (i=0)
X ← ith element of first array
pointer ← pointer + Length
/* pointer points the ith element of second array */
Y ← ith element of second array
pointer ← pointer + Length
/* pointer points the ith element of sum array */
ith element of sum array ← X+Y
pointer ← pointer − 2*Length
/* pointer points the ith element of first array */
pointer ← pointer + 1, [i++]
/* pointer points the i+1 th element of first array */
counter ← counter − 1
if count ≠ 0
True
false
STOP

- Observation Table

Memory Addr	Before	After	
3000 H	05	05	
3001 H	01	01	
3002 H	02	02	
3003 H	03	03	
3004 H	04	04	
3005 H	05	05	
3006 H	08	08	
3007 H	01	01	
3008 H	04	04	
3009 H	02	02	
300A H	00	00	
300B H	00	09	[01+08]
300C H	00	03	[02+01]
300D H	00	07	[03+04]
300E H	00	06	[04+02]
300F H	00	05	[05+00]

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- Code Mnemonic Code Hex Code

	LDA	3000H	2000:	2001: 00	2002: 30
	MOV	B,A	2003: 47		
	MOV	C,A	2004: 4F		
	LXI	H,3001H	2005: 21	2006: 01	2007: 30
back:	MOV	D,M	2008: 56		
	MOV	A,L	2009: 7D		
	ADD	B	200A: 80		
	JNC	skip1	200B: D2	200C: 0F	200D: 20
	INR	H	200E: 24		
skip1:	MOV	L,A	200F: 6F		
	MOV	A,M	2010: 7E		
	ADD	D	2011: 82		
	MOV	D,A	2012: 57		
	MOV	A,L	2013: 7D		
	ADD	B	2014: 80		
	JNC	skip2	2015: D2	2016: 19	2017: 20
	INR	H	2018: 24		
skip2:	MOV	L,A	2019: 6F		
	MOV	M,D	201A: 72		
	MOV	A,L	201B: 7D		
	SUB	B	201C: 90		
	JNC	skip3	201D: D2	201E: 21	201F: 20
	DCR	H	2020: 25		
skip3:	SUB	B	2021: 90		
	JNC	skip4	2022: D2	2023: 26	2024: 20

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	DCR H	2025: 25			
Skip4:	MOV L,A	2026: 6F			
	INR L	2027: 2C			
	JNC skip5	2028: D2	2029: 2C	202A: 20	
	INR H	2028: 24			
skip5:	DCR C	202C: 0D			
	JNZ back	202D: C2	202E: 08	202F: 20	
	RST 5	2030: EF			

- Result: This program is working good for finding the sum of two arrays of N 8bit numbers stored at consecutive memory locations.

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